MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SET AN/ASN-55

1. SCOPE

1.1 This specification covers the requirements for one type of attitude heading reference set, designated AN/ASN-55.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-T-19576	Transmitter, Remote Compass, Thin Wing Type ML-1 (Unstabilized)
MIL-I-22075	Indicator, Bearing-Distance-Heading ID-663/U
MIL-G-25597	Gyroscope, Displacement, Roll and Pitch, Type MD-1
MIL-C-26485	Control Assembly, Attitude Gyroscopic AF/A24G-1
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-H-27269	Horizontal Situation Indicator Group AF/A24J-1
MIL-I-27619	Indicator, Attitude ARU-11/A
MIL-I-27710	Indicator, Attitude ARU-13/A, Remote

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MIL-T-38306	Transmitter, Rate Gyroscope T-970/AJB-7
MIL-G-38330	Gyroscope, Displacement CN-990/ASN-55
MIL-A-38331	Amplifier-Power Supply AM-4080/ASN-55
MIL-G-38332	Gyroscope, Rate Switching CN-1050/ASN-55
MIL-C-38333	Controller, Compass System C-6448/ASN-55

STANDARD

Military

MIL-STD-831	Test Reports, Preparation of

DRAWING

Air Force

64J1546	Schematic Wiring Diagram - Reference Set, Attitude Heading AN/ASN-55

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

McDonnell Aircraft Corporation Specifications

MAC-SCD-32-87120-19	Computer-Airborne Navigation Specification Control Drawing
MAC-SCD-53-87012-301	Navigation Set, Inertial
MAC-SCD-53-87535	Vertical Stabilizer Camera Mount

(Application for copies should be addressed to the McDonnell Aircraft Corporation, Box 516, St. Louis, Missouri 63166.)

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3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The reference set shall consist of the following:

a. Adapter-Compensator, Compass ADK-182/A24G-1A

b. Controller, Compass System C-6448/ASN-55

c. Gyroscope, Rate Switching CN-1050/ASN-55

d. Indicator, Attitude ARU-11/A

e. Transmitter, Rate Gyroscope T-970/AJB-7

f. Amplifier-Power Supply AM-4080/ASN-55

g. Gyroscope, Displacement CN-990/ASN-55

3.3 Associated equipment.- The reference set shall meet all requirements specified herein and shall not be damaged when operated with, or be damaged by operating of, the following equipment. When specified (see 6.2) this equipment will be loaned by the Government to the contractor upon his request.

Item	Type or Part No.	Document
a. Horizontal situation indicator	AF/A24J-1	MIL-H-27269
b. Bearing distance heading indicator	ID-663A/U	MIL-I-22075
c. Flight director computer	Collin P/N 522-1763-003, or equal	-----
d. Inertial navigation set	-----	MAC-SCD 53-87012-301

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Item	Type or Part No.	Document
e. Attitude indicator	ARU-13/A	MIL-I-27710
f. Integrated electronic central	AN/ASQ-88	MIL-I-18309
g. Remote compass transmitter(flux valve)	ML-1	MIL-T-19576
h. Computer-airborne navigation	AN/ASN-46	MAC-SCD-32-87120-19
i. Displacement gyroscope, roll and pitch	MD-1	MIL-G-25597
j. Vertical stabilized camera mount	-----	MAC-SCD 87535

3.4 Design and construction.- The reference set shall be designed and constructed as required by the applicable component specifications.

3.4.1 Reliability.- The reference set shall have a mean-time-between-failures (MTBF) of 375 hours. The reliability test program shall be based upon a discrimination ratio of 0.67 and a confidence level of 90 percent.

3.4.1.1 Longevity.- The reference set shall have an operating life span (equipment longevity as defined in MIL-R-26667) of not less than 2,000 hours before it consistently fails to meet the MTBF index.

3.4.2 Maintainability.- The following maintainability requirements shall be considered in the design of the indicator:

a. Minimization of complexity of maintenance tasks (for example: Calibration, adjustments, inspections, et cetera)

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by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive indentification of the replaceable defective part, assembly, or component

d. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

3.4.3 Interconnection requirements.- Unless otherwise specified, the components of the reference set shall be interconnected and power shall be supplied as shown on AF Drawing 64J1546.

3.4.4 Power.- The power required for starting and operating shall not exceed the following:

a.	115/200V ac 400 cps 3-phase wye	300va
b.	28V dc	20w
c.	5V ac (lighting)	10w
d.	14/28V ac (lighting)	8w.

3.4.5 Attitude flight reference display.- The reference set shall be designed to perform as an all attitude flight reference display. In the primary mode, the reference set shall receive inputs of roll, pitch, and heading from the inertial navigation set. In the backup function to the inertial navigation set, the reference set shall receive roll

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and pitch inputs from the Type MD-1 gyroscope conforming to MIL-G-25597 and shall receive heading signals from the CN-990/ASN-55 gyroscope conforming to MIL-G-38330. A relay shall be provided in the reference set for control of the fast erection cycle of the Type MD-1 gyroscope.

3.5 Performance.- The reference set shall operate within the accuracies specified herein under the following conditions:

a. Attitude: All attitudes except that during standby operation, the CN-990/ASN gyroscope leveling axis shall be limited to "$\pm 82^{\circ}$."

b. Aircraft dynamic response

(1) Aircraft maximum angular rate

(a) Pitch rate: 40°/sec

(b) Roll rate: 300°/sec

(c) Azimuth rate: 40°/sec

(2) Aircraft maximum acceleration

(a) Pitch acceleration: $300^{\circ}/sec^2$

(b) Roll acceleration: $1,500^{\circ}/sec^2$

(c) Azimuth acceleration: $200^{\circ}/sec^2$

c. Acceleration

(1) Up: 3.3g's

(2) Down: 9g's

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(3) Forward: 6g's

(4) Aft: 2g's

(5) Lateral: 4g's

3.5.1 Fast erection cycle.- An automatic circuit shall be provided for the fast erection cycle of the Type MD-1 gyroscope. The cycle shall be initiated by switching to the SYNC position on the C-6448/ASN-55 controller conforming to MIL-C-38333. The cycle shall maintain the Type MD-1 gyroscope in the fast erection mode for a minimum of 45 seconds.

3.5.2 Input signals.- During the primary mode of operation, the reference set shall accept from the inertial navigation set roll and pitch input signals for display on the ARU-11/A attitude indicator conforming to MIL-I-27619, and true aircraft heading for the stabilization input of the ADK-182/A24G-1A adapter-compensator conforming to MIL-C-26485. Switching of the inertial inputs shall be performed in the aircraft relay panels. The Type ML-1 remote compass transmitter (flux valve) shall supply the reference signal for SLAVE mode.

3.5.3 Output signals.- The reference set shall provide the output signals listed in table I for use by other aircraft equipment. All synchro outputs shall be excited by the aircraft phase-C power.

TABLE I

OUTPUT SIGNALS

Output Signal No.	Output Signal	Load	Excitation
1	Magnetic heading	2 torque diff. transmitter, Clifton CDC-8-A-1, or equal	Phase C 26V
2	Magnetic heading	Same as No. 1	Phase C 26V

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Output Signal No.	Output Signal	Load	Excitation
3	Magnetic heading	4Ct's, Clifton CTC-8-A-4, or equal	Phase C 26V
4	Magnetic heading	Same as No. 3	Phase C 26V
5	Turn rate switching	3K ohm impedance	Phase A 115V
6	PRIM/STBY selector	-----	28V

3.5.3.1 Accuracy.- The reference set output signals shall meet the accuracy requirements specified in table II. Dynamic accuracies shall be met under any combination of the dynamic conditions specified in 3.5b(1) for a 5-minute period. The reference set conditions specified in 3.5b(2) shall be met for a period of 30 seconds.

TABLE II

OUTPUT SIGNAL ACCURACY

Signal Output	Error (maximum-3 sigma)	
	Static	Dynamic
Roll angle (indicator only)	0.5°	10°
Pitch angle (indicator only)	0.5°	2°
Magnetic heading	0.75°	2.5°
DC drift	6°/hr	2°
Turn rate indicator	1°/min	9°/min (limited angle rate capability)

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3.5.3.1.1 Standby operation.- During standby operation, the heading errors specified in table II shall apply for pitch and roll angles of less than 10°.

3.5.4 Radio noise interference.- The reference set shall meet the radio noise interference requirements of MIL-I-6181.

3.6 Controller.- The C-6448/ASN-55 controller shall provide switching from primary (PRIM)(i.e. inertial navigational set) to standby (STBY) attitude systems. It shall also provide switching to the compass (COMP) directional gyro (DG) slaved (SLAVED), and synchronizing (SYNC) modes of operation.

3.7 Displacement gyroscope.- The CN-990/ASN-55 gyroscope shall provide a signal for azimuth stabilization to the ADK-182/A24G-1A adapter-compensator during standby operation.

3.8 Amplifier-power supply.- The amplifier shall conform to MIL-A-38331. The amplifier shall contain the servo amplifier for leveling the inner gimbal of the displacement gyroscope, the WYE-DELTA transformer, and the fast erection cycle control relay for the Type MD-1 gyroscope.

3.9 Adapter-compensator.- The ADK-182/A24G-1A adapter-compensator shall operate in conjunction with the C6448/ASN-55 controller. These two components, along with the CN-990/ASN-55 displacement gyroscope and the Type ML-1 transmitter, shall develop gyro-stabilized magnetic (SLAVED), direct-reading magnetic (COMP), and free gyro (DG) heading for the attitude indicator and other aircraft equipment. Means shall be provided for compensating extraneous magnetic effects peculiar to aircraft.

3.10 Rate switching gyroscope.- The CN-1050/ASN-55 rate switching gyroscope shall conform to MIL-G-38332. It shall prevent erection of the Type MD-1 gyroscope to a false gravity vertical and shall prevent improper slaving of the

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CN-990/ASN-55 gyroscope by removing erection and slaving voltages within the time intervals and turn rates specified in MIL-G-38332.

3.11 Indicator.- The attitude indicator shall present an unambiguous display of aircraft attitude throughout all maneuvers. The display shall be scalar indexed to indicate the degree of displacement in pitch, roll, and azimuth.

3.12 Transmitter.- The transmitter shall conform to MIL-T-38306. The transmitter shall provide a d-c signal, proportional to the aircraft rate of turn, to the attitude indicator for display.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of components of the system shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Interconnections and power.- Unless otherwise specified, the components of the reference set shall be interconnected and power shall be applied as shown on AF Drawing 64J1546. A relay operated by J801-E shall

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switch J901-r,-s, -t from J1201-N, -R, and -T to a synchro transmitter excited by phase C. The synchro transmitter shall be connected to J1502-a, -b, and -c and J502-e, -f, and -g. The Type MD-1 gyroscope will not be required when the synchro transmitter is used.

4.3.2 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C).

4.3.3 Attitude.- Except as specified below, the system components shall be tested in the normal operating position.

a. Unless otherwise specified, the CN-990/ASN-55 gyroscope shall be secured to a mounting surface which can be leveled within 0.1° and which can be rotated 360° about the azimuth axis.

b. When specified, the CN-990/ASN-55 gyroscope shall be secured to the surface of a Scorsby table which can be leveled within 0.1° and which can be oscillated simultaneously about three mutually perpendicular axes. The amplitude of each node of oscillation shall be 3° and the frequency shall be 5 to 7 oscillations per minute

c. The CN-1050/ASN-55 rate switching gyroscope shall be mounted on a rate table which can be rotated at a rate of 15° per minute about an axis perpendicular to the mounting surface

d. The T-970/AJB-7 transmitter shall be mounted on a rate table which can be rotated 360° ±10° per minute.

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4.3.4 Special equipment

4.3.4.1 Flux valve signal source.- The flux valve simulator shall be as shown on figure 1 or shall have equivalent outputs. Unless otherwise specified, the field strength shall be set at normal. The components shall be as follows:

C1	0.925 uf ±2 percent
CR1 through CR4	1N645
R1	312 ohms ±2 percent
R2	33.4 ohms ±2 percent
R3	17.0 ohms ±2 percent
R4	13.4 ohms ±2 percent
Z_x	1009 ±5 percent + j1194 ±5 percent ohms @800 cps sine wave.
B1	Bendix AY 500-25 synchro, or equivalent
R5	Selected to provide a circuit load at J902-p to q of 0.044 ±0.002 amp at 23.5V

Maximum wattage of any resistor is 0.5.

4.3.4.2 Servoed indicator.- In tests which require a more precise measurement of angular displacement than can be realized with the ARU-11/A indicator, it will be permissible to use a suitable servoed indicator. The servoed indicator shall be connected in parallel across the appropriate synchro transmission leads.

4.3.5 Input power.- The following power inputs shall be connected as shown on AF Drawing 64J1546.

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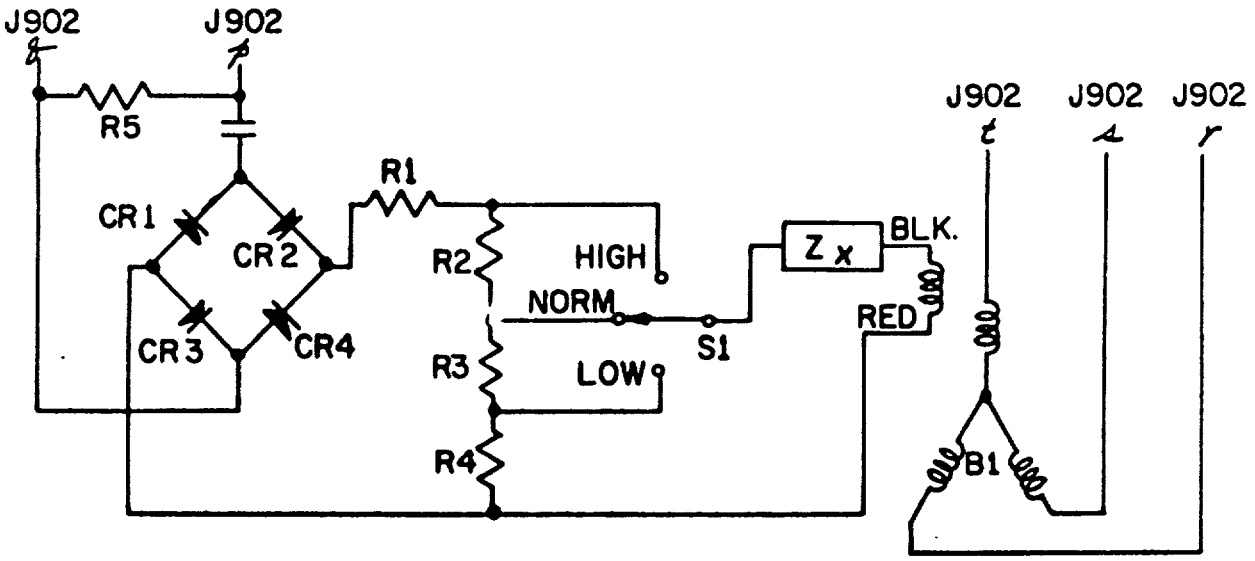


FIGURE 1, FLUX VALVE SIGNAL SOURCE

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a. 200V ac rms ±2 percent line-to-line, 115V ac rms ±2 percent line-to-neutral, 400 cps ±1 percent 3-phase, 4-wire WYE, with neutral grounded

b. 26V ac ±2 percent 400 cps ±1 percent single phase

c. 27.5 ±1V dc

d. 5 ±0.2V ac or dc

4.3.6 Preliminary adjustments.- The following adjustments and control settings shall be made prior to testing and shall remain at these settings unless otherwise specified.

a. The mode selector switch on the compass system controller shall be set to the SLAVED position

b. The latitude compensation potentiometer on the compass system controller shall be set to the latitude at the test site.

c. The PRIM-STBY switch on the compass system controller shall be set to the STBY position

d. The bias correction dial on the compass adapter shall be set to zero.

e. The deviation compensator on the compass adapter shall be adjusted for heading errors

f. The flux valve signal source (4.3.4.1) shall be set to 0° heading

g. The pitch trim control of the attitude indicator shall be adjusted for zero trim

h. The pitch and roll simulators shall be set to 0°.

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4.3.7 System interconnection.- Each component of the reference set shall be connected to the system interconnect panel as shown on figure 2.

4.4 Preproduction testing

4.4.1 Test samples.- The test samples shall consist of 9 reference sets representative of the production equipment. Three of the reference sets shall be subjected to all the tests except reliability and the other 6 reference sets shall be subjected to the reliability test only. The samples, identified with the manufacturer's part number and such other information as required, shall be submitted to the procuring activity.

4.4.1.1 Reliability demonstration.- Unless otherwise specified, 6 reference sets shall be selected at random from the first 50 produced on the contract or order and subjected to the test specified in 4.6.15 for a total of 12,375 hours unless rejection is determined in accordance with figure 3 prior to that time.

4.4.2 Test report.- A test report shall be prepared in accordance with MIL-STD-831 and two complete copies of the report shall be forwarded to the procuring activity. The report shall include the results of the individual tests conducted under the applicable component specifications.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Acceptance tests.- Acceptance tests shall consist of the sampling tests.

4.5.1 Sampling tests

4.5.1.1 Sampling plan A.- The following number of samples selected at random from the number specified on the contract or order, shall be subjected to the tests listed below and described under 4.6.

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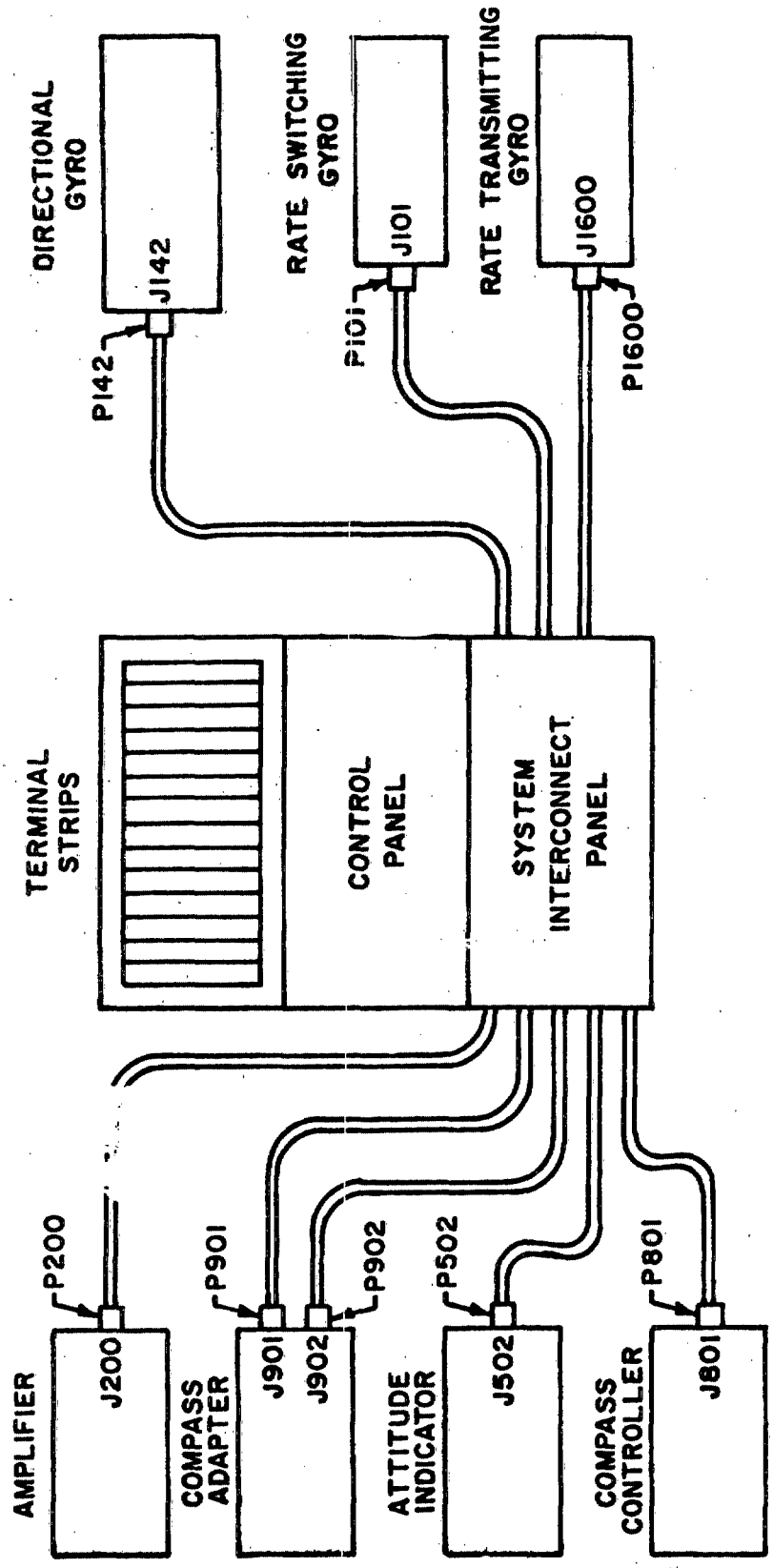


FIGURE 2. AN/ASN-55 SYSTEM INTERCONNECT

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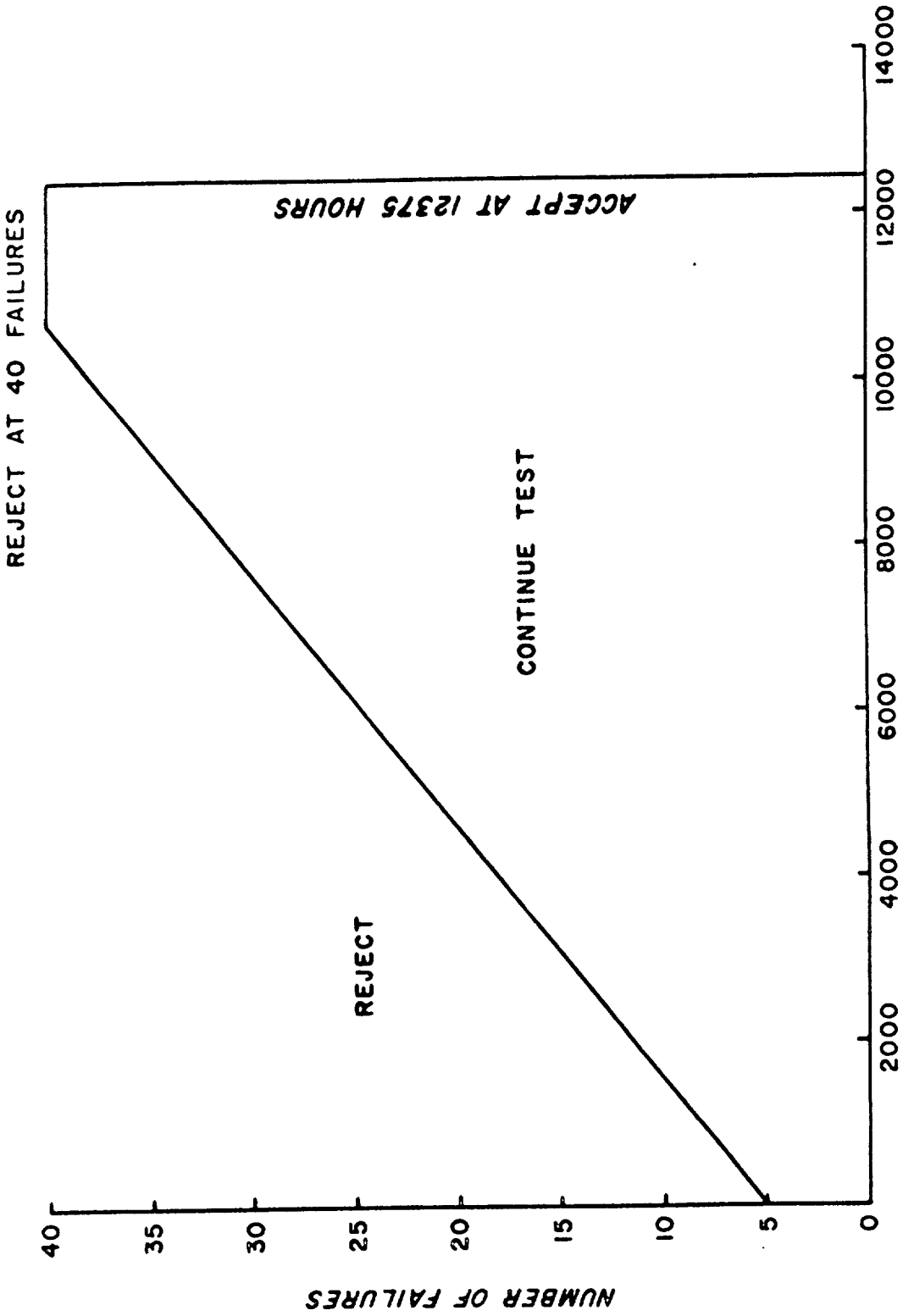


FIGURE 3. REJECT-CONTINUE TEST CRITERIA

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Quantity Produced	No. of Samples
1-15	3
16-100	1
101 and over	1 per 100 or major portion thereof

a. Start cycle

b. Azimuth performance

c. Yaw rate switching

d. Mode switching synchronization

e. Primary-standby synchronization

f. Manual synchronization and fast erection

g. Servo sensitivity

h. Set heading slew

i. Latitude compensation drift

j. Turn rate indication

k. Compass mode operation

l. Radio noise interference

m. Power consumption.

4.5.1.2 Rejection and retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failures are determined.

4.5.1.3 Corrective action and retest.- Corrective action and retest shall be required for each failure.

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4.5.2 Defects in items already accepted.- If the investigation of a test failure indicates that defects may exist in items already accepted, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5.3 Sampling plan B.- Unless otherwise specified, 4 reference sets shall be selected at random from each 50 produced on the contract or order and subjected to the test specified in 4.6.15 for a total of 2,475 hours.

4.5.3.1 Acceptance or Rejection.- Figure 4 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.15. In the event the continue-test decision applies after 2,475 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept-reject decision. If the continue-test condition persists, the data from successive production quantities shall be accumulated until 40 failures or 12,375 hours of testing have occurred, at which time an accept-reject decision shall be made.

4.6 Test methods

4.6.1 Start cycle.- The power specified in 4.3.5 shall be applied. Within 90 seconds after power application, the following shall occur:

a. The OFF flag of the attitude indicator shall move out of view

b. The azimuth heading indication of the attitude indicator shall repeat the setting of the flux valve signal source within 2.5^{o}

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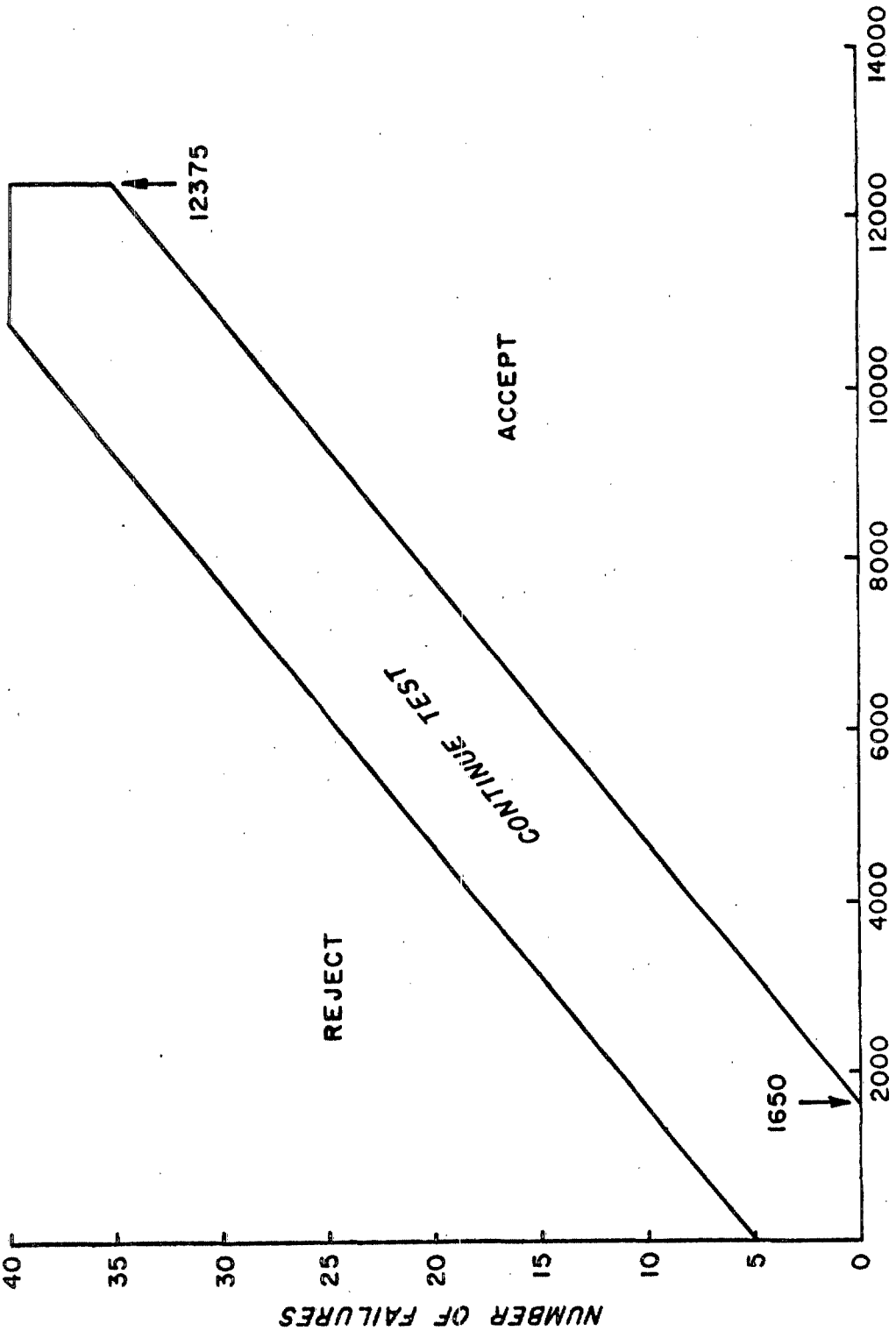


FIGURE 4. ACCEPT-CONTINUE TEST-REJECT CRITERIA

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c. The sync meter of the compass system controller shall indicate null ±1/4 of full scale

4.6.1.1 Power failure warning.- Phase C power shall be disconnected. The OFF flag on the attitude indicator shall move into view immediately. Phase C shall be re-applied and the OFF flag shall move out of view within 3 seconds.

4.6.2 Azimuth performance.- The mode select switch on the compass controller shall be set to the DG position. The displacement gyroscope shall be rotated through 360° in azimuth at a rate of approximately 2 rpm. The azimuth indication of the attitude indicator shall follow smoothly within ±2.25° accuracy and shall not indicate hunting or jumping in azimuth, roll, or pitch in excess of 0.5°. A suitable servoed indicator shall be connected to pins a,b, and c of J902 and the above test repeated. This test shall be performed for both directions of azimuth.

4.6.2.1 The mode selector switch on the compass controller shall be set to the SYNC position, then to the DG position. With recording equipment connected, the displacement gyroscope shall be rotated in azimuth at a rate of 40° per second. The attitude indicator azimuth servo position shall be measured and shall not lag the rate table position by more than 3.0°. This test shall be performed for both directions of azimuth.

4.6.3 Yaw rate switching.- The mode selector switch shall be positioned to SLAVED. The set heading switch shall be used to slew the azimuth heading indication 30° from the flux valve signal source position. The rate switching gyroscope shall be mounted as in 4.3.3c and rotated at a rate of 15° per minute. Within 30 seconds of rotation, 115V ac shall be measured on J200-M with respect to ground. It shall be in phase with the voltage at J200-b to J200-a. While rotation is continued, the mode selector switch shall be moved to the SYNC position and released. The azimuth heading indication shall not synchronize with the flux valve signal source.

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The rotation of the rate switching gyroscope shall be stopped. After 30 seconds, the mode selector switch shall be moved to SYNC. The azimuth indication shall synchronize to the flux valve signal source $\pm 2.5^{\circ}$.

4.6.4 Mode switching synchronization.- The mode selector switch shall be moved to the DG position. The set heading switch shall be operated to slew the azimuth indication to a position 120° from the flux valve signal source setting. The mode selector switch shall be moved to the SLAVED position. The azimuth heading indication shall synchronize to the flux valve signal source $\pm 2^{\circ}$ within 5 seconds.

4.6.5 Primary-standby synchronization.- The set heading switch shall be operated to slew the azimuth indication 120° from the flux valve setting. The PRIM-STBY switch on the compass controller shall be moved to the PRIM position. The azimuth heading indication shall synchronize to the flux valve signal source. The heading indication shall again be slewed 120° from the flux valve signal source setting. The PRIM-STBY switch shall be returned to the STBY position, and the heading indication shall again synchronize to the flux valve signal.

4.6.6 Manual synchronization and fast erection.- The set heading switch shall be operated to slew the heading indication 120° from the flux valve signal source. The mode selector switch shall be moved to the SYNC position and released. The heading indication shall synchronize to the flux valve signal source $\pm 2^{\circ}$ within 5 seconds. Switching to the SYNC position shall also cause 115V ac to be measured on J200-N with respect to J200-a. The duration of the voltage shall be not less than 45 nor more than 80 seconds.

4.6.7 Servo sensitivity.- The displacement gyroscope shall be rotated 0.5° about the heading axis (an axis perpendicular to its mounting surface). There shall be a perceptible movement of the heading indication of the attitude indicator.

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4.6.8 Set heading slew.-The set heading control shall be rotated clockwise. The azimuth heading shown on the attitude indicator shall slew up-scale. The rate of slew shall increase proportionately with clockwise rotation of the set heading control. With the set heading control in the extreme clockwise position, the rate of slew shall be 360° in not less than 7.5 seconds. Return the set heading control toward its center position. It shall be possible to maintain a continuous slewing rate which does not change the azimuth heading shown on the attitude indicator more than 30° in 10 seconds. With the set heading control displaced in the counterclockwise direction, the azimuth heading shown on the attitude indicator shall slew down-scale. The rate of slew shall be 360° in not less than 7.5 seconds.

4.6.9 Latitude compensation drift.- Prior to starting this test, the reference set shall be operated for at least 15 minutes. The displacement gyroscope shall be operated on a Scorsby table as specified in paragraph 4.3.3b. The latitude dial of the compass controller shall be set for the local latitude and the mode selector switch shall be in the DG position. The heading indication shall be observed before and after a 15-minute interval. The heading indication should not change more than 2°. If a change of more than 2° is observed, the interval shall be extended to 1 hour. The heading change during the 1-hour interval shall be not more than 7°.

4.6.10 Turn rate indication.- The rate gyroscope transmitter shall be mounted as specified in 4.3.3d and rotated in a clockwise direction as viewed from the top until the rate indicator in the attitude indicator is deflected two scale divisions to the observer's right. To achieve this condition, the velocity of the rate table shall be 1 rpm $\pm$14 percent.

4.6.11 Compass mode operation.- The mode selector switch on the compass controller shall be set to the COMP position. The flux valve signal source shall be rotated through 360° in 30° increments. In each case, the azimuth heading shown on the attitude indicator shall agree with the heading set in on the flux valve signal source within 2°. The displacement gyroscope shall be rotated about the heading axis. The heading display shall not follow.

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4.6.12 Radio noise interference.- The radio noise interference test shall be conducted in accordance with MIL-I-6181.

4.6.13 Power consumption.- With the system connected as specified in 4.3.1, the power consumption shall not exceed the following:

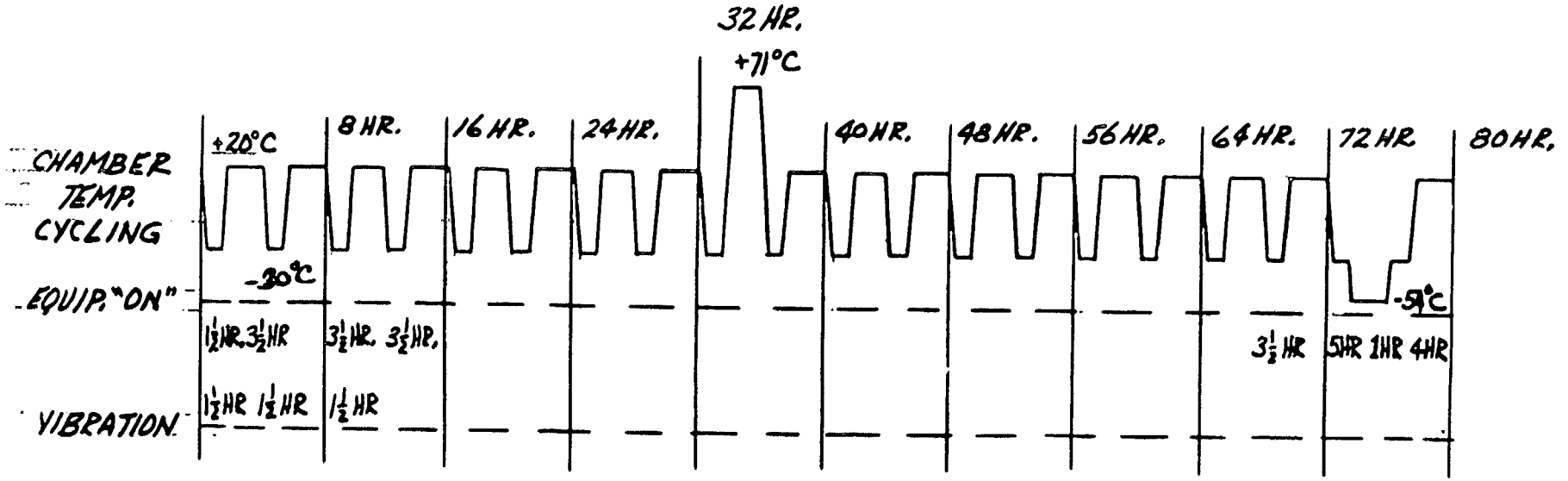
115/200V ac 400 cps	3 phase, WYE	300va
28V dc		20w
5V ac (lighting)		10w
14/28V ac (lighting)		8w

4.6.14 Associated equipment compatibility.- Compatibility tests as required to demonstrate compliance with 3.3 shall be conducted.

4.6.15 Reliability.- The components shall be properly connected as a system as shown on AF Drawing 64J1546 and operated to assure proper set up. The reference set, less compass controller and attitude indicator, shall be placed in test chambers which meet the environmental cyclic conditions shown on figure 5. During the ON cycle, the reference set shall be exercised by a 3-axis, 15° total movement, 5 to 7 cpm, reversing scorsby action applied to the CN-990/ASN-55 displacement gyroscope and by the vibration shown on figure 5. The rate switching gyroscope and the rate gyroscope transmitter shall be mounted on a Genisco rate table. The accelerometer shall also be mounted on a Genisco rate table that is capable of producing 4g's acceleration along the sensitive axis of the accelerometer. After each 160-hour cycle of operation, the reference set shall meet the following:

4.6.15.1 Application of power

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NOTES:

a. Environment cycle repeats every 80 hours.

b. Temp rate of change of chamber shall be 0.1° - 0.5° C per second

c. Vibration level 0.2g at 20 to 30 cps

d. OFF periods for equipment ON and OFF are 1/2 hour (except 76th hour)

e. Monitoring of system outputs for catastrophic type failures performed during each ON cycle

f. Satisfactory performance tests on systems individual units conducted every 160 hours of environment cycle to detect performance deviation type failures.

FIGURE 5 RELIABILITY TEST ENVIRONMENT CYCLES

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4.6.15.1.1 Start cycle.- Power shall be applied to the reference set by placing the proper switches of the system interconnect panel in the ON position. The starting sequence shall be completed in less than 80 seconds after application of power. The start cycle will be completed when the attitude indicator OFF flag disappears from view.

4.6.15.1.2 Power failure.- Power to the reference set shall be interrupted by removing phase B or phase C fuse from the system. The attitude indicator OFF flag shall appear when power is interrupted.

4.6.15.2 Attitude display

4.6.15.2.1 Heading synchronization.- Within 1.5 minutes after the application of power, the indicator shall repeat the flux valve simulator setting within 3° and the sync indicator shall indicate within 1/4 of full scale from null. Within 3.5 minutes, the attitude indicator shall repeat the flux valve within 1.5° and the sync indicator shall indicate null.

4.6.15.2.2 Final erection.- Three minutes after the application of power, the attitude indicator shall indicate level within 1.5° in both the pitch and roll axes.

4.6.15.2.3 Roll performance.- The roll simulator on the system interconnect panel shall be turned through 360° in roll at a rate of approximately 2 rpm. The attitude indicator shall follow this displacement smoothly, and shall not jump in pitch, roll, or azimuth. This procedure shall be repeated with the direction of rotation reversed.

4.6.15.2.4 Pitch performance.- The pitch simulator on the system interconnect panel shall be turned through 360° in pitch at a rate of approximately 2 rpm. The attitude indicator shall follow this displacement smoothly, and shall not jump in pitch, roll, or azimuth. This procedure shall be repeated with the direction of rotation reversed.

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4.6.15.2.5 Azimuth performance.- The COMP-DG-SLAVED-SYNC switch on the compass controller shall be set to the DG position. The directional gyroscope shall be rotated through 360° in azimuth at a rate of approximately 2 rpm. The attitude indicator shall follow this displacement smoothly and shall not jump in azimuth, roll, or pitch. This procedure shall be repeated with the direction of rotation reversed. The switch shall then be set to the SLAVED position.

4.6.15.2.6 Yaw rate switching.- The attitude indicator's azimuth indication shall be slewed approximately 30° with the push-to-turn-switch on the compass controller. The Genisco rate table shall be adjusted so that the switching rate gyroscope will rotate at a rate of 15° per minute. After the gyroscope has rotated for 35 seconds, maximum, the roll erection cutout bulb on the system interconnect panel shall light. The controller mode switch shall then be set to the SYNC position. The azimuth indication shall not synchronize to the flux valve simulator in the system interconnect panel. The Genisco rate table shall be stopped. After 20 seconds, the mode switch shall be turned to the SYNC position, the azimuth indication shall then synchronize with the flux valve simulator, and the roll erection cutoff light shall go out.

4.6.15.2.7 Pitch trim.- Rotation of the attitude indicator pitch trim knob to its two extreme positions shall cause a minimum indication change of 5° in climb and 10° in dive (cw).

4.6.15.2.8 Switch-Over Sychronization.- Set the mode switch on the Compass Controller to the DG position. A 170° heading shall be set in on the compass transmitter simulator. The mode switch shall be returned to the SLAVED position. The azimuth heading shown on the attitude indicator shall slew to within 2° of the 170° heading set on the compass transmitter simulator within 5 seconds maximum and the sync indicator shall approach a null position.

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4.6.15.2.9 Manual synchronization.- With the azimuth heading shown on the attitude indicator synchronized to the 170° heading set in on the compass transmitter simulator, a 0° heading shall be set in on the compass transmitter simulator. The mode switch on the compass controller shall be turned to the SYNC position and then released. The azimuth heading shown on the attitude indicator shall synchronize to within 2° of the 0° heading set in on the compass transmitter simulator within 5 seconds, maximum, and the sync indicator shall approach a null position.

4.6.15.2.10 Sync meter.- With the azimuth heading shown on the attitude indicator synchronized to the 0° heading set in on the compass transmitter simulator, the compass transmitter simulator shall be rotated to a 3° heading. The sync indicator shall deflect to the observer's left (+).

4.6.15.2.11 Servo sensitivity.- There shall be perceptible movement of the ARU-11/A attitude indication when the directional gyroscope is displaced a maximum of 1° from 0° in azimuth.

4.6.15.2.12 Azimuth slew.- the attitude indicator azimuth indication shall increase when the compass controller's push to turn knob is depressed and rotated clockwise. Counter clockwise rotation of the push to turn knob shall produce a decreasing azimuth indication.

4.6.15.2.12.1 Azimuth performance.- The COMP-DG-SALVED-SYNC switch on the compass controller shall be set to the DG position and the directional gyroscope rotated through 360° in azimuth at a rate of approximately 2 rpm. The azimuth indication of the attitude indicator shall follow smoothly with $\pm$2.25° accuracy and shall not indicate hunting or jumping in azimuth, roll, or pitch in excess of 0.5°. A suitable servoed indicator shall be connected to pins a,b, and c of J902 and the above test repeated. This test shall be performed for both directions of azimuth.

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4.6.15.2.12.2 The COMP-DG-SLAVED-SYNC switch on the compass controller shall be set to the SYNC position and then to the DG position. With the required recording equipment properly connected, the directional gyroscope shall be rotated in azimuth at a rate of 40° per second. The attitude indicator azimuth servo position shall be measured and shall not lag the rate table position by more than 2.25°. This test shall be performed in both directions of azimuth.

4.6.15.2.13 Slaving.- The displacement gyroscope shall be rotated a few degrees in azimuth to displace the compass controller's sync indicator from null but not to full deflection. A detectable motion of the sync indicator toward null shall be observed. This procedure shall be repeated for displacements in the opposite direction.

4.6.15.2.14 The directional gyroscope shall be rotated in azimuth until a null voltage is obtained between pins M and L of J142. The directional gyroscope shall be rotated slowly (approximately 2 rpm) through 360° about the pitch axis. The change of azimuth indication on the attitude indicator shall be recorded. With a null voltage between pins M and L of J142, the gyroscope shall be rotated slowly (approximately 2 rpm) through 360° in roll. The change of azimuth indication on the attitude indicator shall be recorded. A period of 5 minutes shall be allowed for the gyroscope to stabilize before proceeding with subsequent tests.

4.6.15.2.15 Directional gyroscope drift.- The ARU-11/A azimuth indication shall be observed for a period of 15 minutes with the system operating in the DG mode and latitude adjustment of the compass controller at zero. The drift (measured on the servoed repeater), corrected for the effects of earth rotation, shall not exceed 6.0° during this period.

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4.6.15.2.16 Turn rate indication.- The rate transmitting gyro shall be rotated on a Genisco rate table at a rate sufficient to cause the turn rate indicator on the attitude indicator to align with the second index to the right of center. The applied rate shall be CW at $6^{o} \pm 1.0^{o}$ per second. This test shall be repeated in the opposite direction, aligning the indicator with the second index to the left of center.

4.6.15.2.17 Compass mode.- Set the switch on the compass controller to the COMP position. The attitude indicator shall not respond to movement of the directional gyro, but will respond in azimuth indication to displacements of the flux valve simulator.

4.6.15.3 Longevity.- At the completion of the reliability test specified in 4.6.15 through 4.6.15.2.17, three of the reference sets shall continue to be tested for a total of 2,000 hours each.

4.6.15.4 Relevancy criteria.- All failures shall be classified as relevant unless approved for exclusion by the procuring activity.

4.6.15.5 Test information.- Sixty days prior to starting the reliability test, the contractor shall submit detailed information to the procuring activity for approval. This information shall include, but not necessarily be limited to:

a. Equipment to be used in monitoring performance

b. Periodicity of calibration checks to made on monitoring equipment

c. Description of environmental test facilities to be used, including recording facilities, method of control of heating and cooling facilities, and vibration equipment

d. Frequency of monitoring.

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4.6.15.6 Test records.- Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-R-26667: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.15.7 Repair of the failed units.- In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.16 Maintainability verification.- Maintainability verification shall be in accordance with MIL-M-26512, Appendix A, paragraph entitled "Maintenance Task Test and Demonstration." All tasks selected shall be effectively simulated.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instruction of procuring activity.

6. NOTES

6.1 Intended use.- The AN/ASN-55 attitude heading reference set covered by this specification is intended for use in the RF-4C aircraft for providing continuous information about the pitch, roll, and azimuth. The reference set is designed for all-weather operations.

6.2 Ordering data.- Procurement documents should specify the following:

a. Type, number, and date of this specification

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b. Associated equipment required

c. Quantity of associated equipment (see 3.3) to be provided

d. When sampling plan B test will not be conducted

e. Levels of packaging and packing required.

6.3 Definition

6.3.1 Failure.- Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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